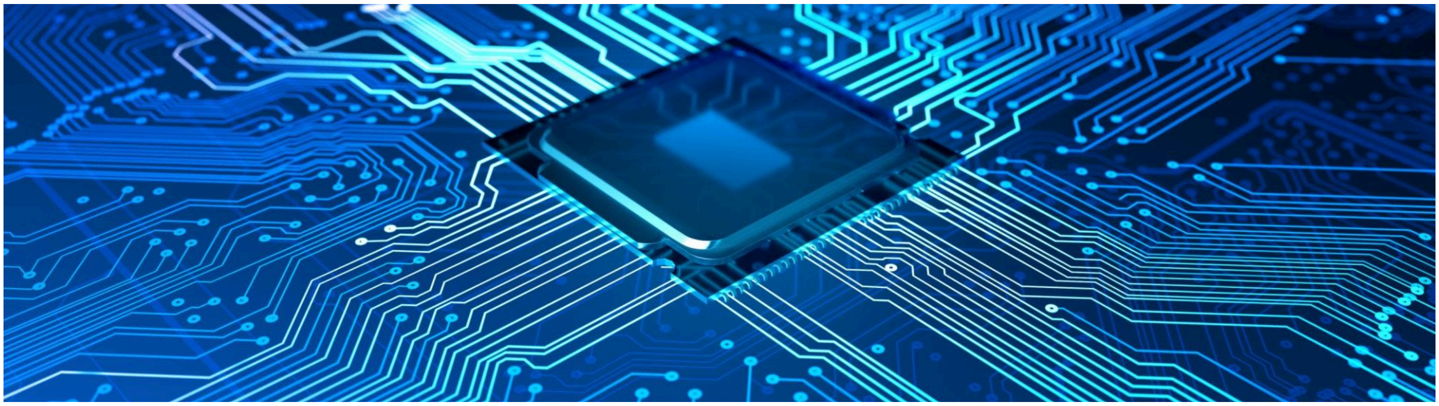
Overview



Mini Project: 🛒 Shopping Cart (Arrays + Objects)

Library book tracker

Ramachandra prakash M

ramachandraprakashvnr@gmail.com

GOAL

- Use Vibe Coding
- Store a list of items in an **array**.
- Each item is an **object**: `{ name, qty, price }`.
- Print a small receipt with template strings.

MY PROMPT

Use vibe coding to store a list of items in an array, where each item is an object with `{ name, qty, price }`. Print a receipt using template strings.

JS code

```javascript
// Ramprakash's AFM JavaScript Project: Shopping Cart

// Step 1 & 2: Create objects for supermarket items
const item1 = { name: "Apple", qty: 3, price: 0.5 };
const item2 = { name: "Bread", qty: 1, price: 2.0 };
const item3 = { name: "Milk", qty: 2, price: 1.5 };
const item4 = { name: "Eggs", qty: 12, price: 0.1 };

// Step 3 & 4: Create and populate the cart array
let cart = [item1, item2, item3, item4];

// Step 5: Print the cart (receipt)
console.log("--- RECEIPT ---");
cart.forEach(item => {
  console.log(`${item.name} x ${item.qty} = $${(item.qty * item.price).toFixed(2)}`);
});

// Calculate and print total cost
const total = cart.reduce((sum, item) => sum + (item.qty * item.price), 0);
console.log(`----------------`);
console.log(`Total: $${total.toFixed(2)}`);
```

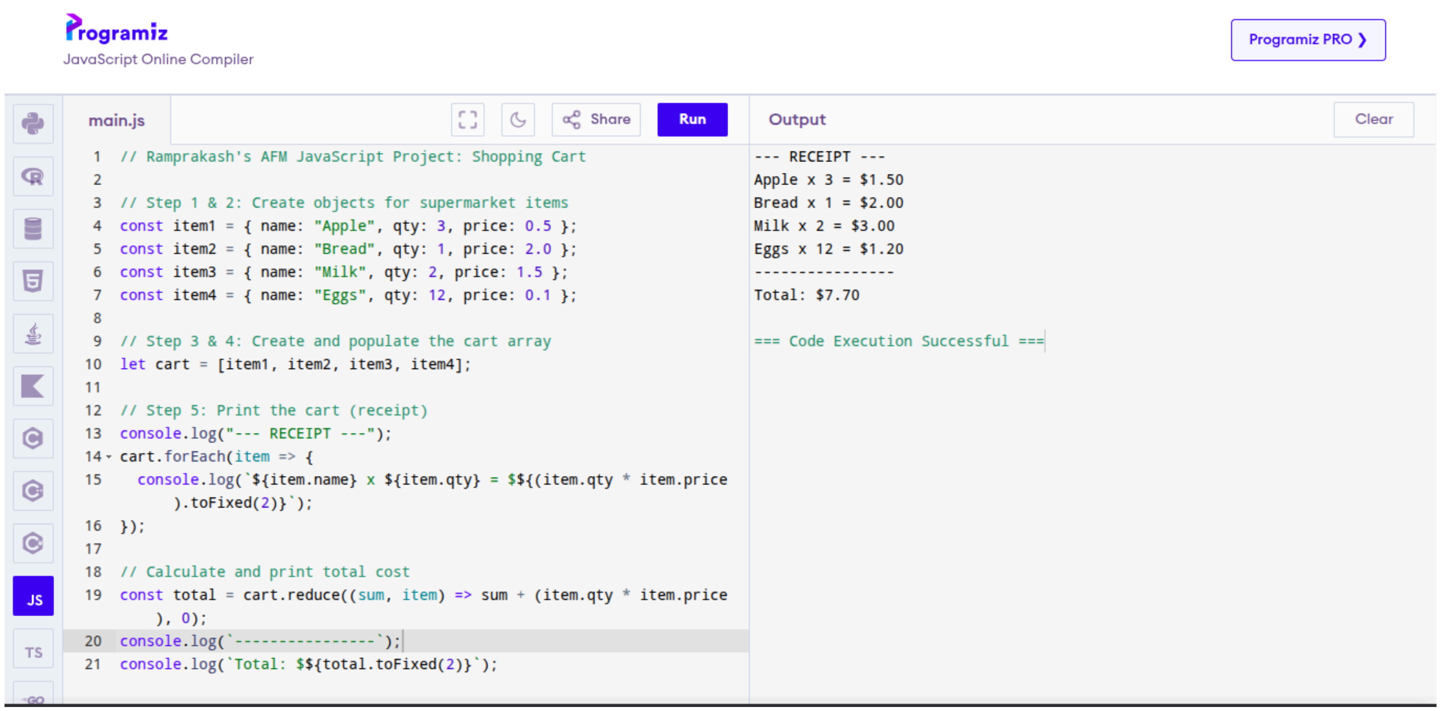
OUTPUT

--- RECEIPT ---
Apple x 3 = $1.50
Bread x 1 = $2.00
Milk x 2 = $3.00
Eggs x 12 = $1.20

Total: $7.70

=== Code Execution Successful ===

SCREENSHOTS



Bonus Mini Project:
📚 Library Book Tracker

Goal

Create a program to track books in a library using arrays and objects. Each book will have properties like `title`, `author`, `isCheckedOut`, and `dueDate` (if checked out). You'll also write functions to check books in/out and list available books.

PROMPT

1. Create an array called library to store book objects.
2. Each book object should have:
   - title (string)
   - author (string)
   - isCheckedOut (boolean, default: false)
   - dueDate (string, default: "" or null)
3. Write functions to:
   - Check out a book (set isCheckedOut to true and add a dueDate).
   - Return a book (set isCheckedOut to false and clear dueDate).

- List all available books (not checked out).

4. Print the library status before and after checking out/returning books.

MY JSCODE

```
// Ramprakash's Bonus Mini Project: Library Book Tracker

// Step 1: Create the library array with book objects
let library = [
  { title: "The Great Gatsby", author: "F. Scott Fitzgerald", isCheckedOut: false, dueDate: "" },
  { title: "To Kill a Mockingbird", author: "Harper Lee", isCheckedOut: false, dueDate: "" },
  { title: "1984", author: "George Orwell", isCheckedOut: false, dueDate: "" }
];

// Step 2: Function to check out a book
function checkOutBook(bookTitle, dueDate) {
  const book = library.find(b => b.title === bookTitle);
  if (book && !book.isCheckedOut) {
    book.isCheckedOut = true;
    book.dueDate = dueDate;
    console.log(`✅ "${bookTitle}" has been checked out. Due: ${dueDate}`);
  } else if (book && book.isCheckedOut) {
    console.log(`❌ "${bookTitle}" is already checked out. Due: ${book.dueDate}`);
  } else {
    console.log(`❌ "${bookTitle}" not found in the library.`);
  }
}

// Step 3: Function to return a book
function returnBook(bookTitle) {
  const book = library.find(b => b.title === bookTitle);
  if (book && book.isCheckedOut) {
    book.isCheckedOut = false;
    book.dueDate = "";
    console.log(`✅ "${bookTitle}" has been returned.`);
  } else if (book && !book.isCheckedOut) {
    console.log(`❌ "${bookTitle}" is not checked out.`);
  } else {
    console.log(`❌ "${bookTitle}" not found in the library.`);
```

```
  }
}

// Step 4: Function to list available books
function listAvailableBooks() {
  console.log("\n--- Available Books ---");
  const availableBooks = library.filter(b => !b.isCheckedOut);
  if (availableBooks.length === 0) {
    console.log("No books available.");
  } else {
    availableBooks.forEach(book => {
      console.log(`- ${book.title} by ${book.author}`);
    });
  }
}

// Step 5: Print initial library status
console.log("--- Initial Library Status ---");
library.forEach(book => {
  console.log(
    `${book.title} by ${book.author} | ` +
    `Checked Out: ${book.isCheckedOut} | ` +
    `Due: ${book.dueDate || "N/A"}`
  );
});

// Step 6: Test the functions
checkOutBook("1984", "2026-05-20");
checkOutBook("The Great Gatsby", "2026-05-15");
checkOutBook("1984", "2026-05-20"); // Try to check out again
listAvailableBooks();

returnBook("1984");
listAvailableBooks();

// Step 7: Print final library status
console.log("\n--- Final Library Status ---");
library.forEach(book => {
  console.log(
    `${book.title} by ${book.author} | ` +
```

```
    `Checked Out: ${book.isCheckedOut} | ` +
    `Due: ${book.dueDate || "N/A"}`
  );
});
```

OUTPUT

--- Initial Library Status ---
The Great Gatsby by F. Scott Fitzgerald | Checked Out: false | Due: N/A
To Kill a Mockingbird by Harper Lee | Checked Out: false | Due: N/A
1984 by George Orwell | Checked Out: false | Due: N/A
✅ "1984" has been checked out. Due: 2026-05-20
✅ "The Great Gatsby" has been checked out. Due: 2026-05-15
❌ "1984" is already checked out. Due: 2026-05-20

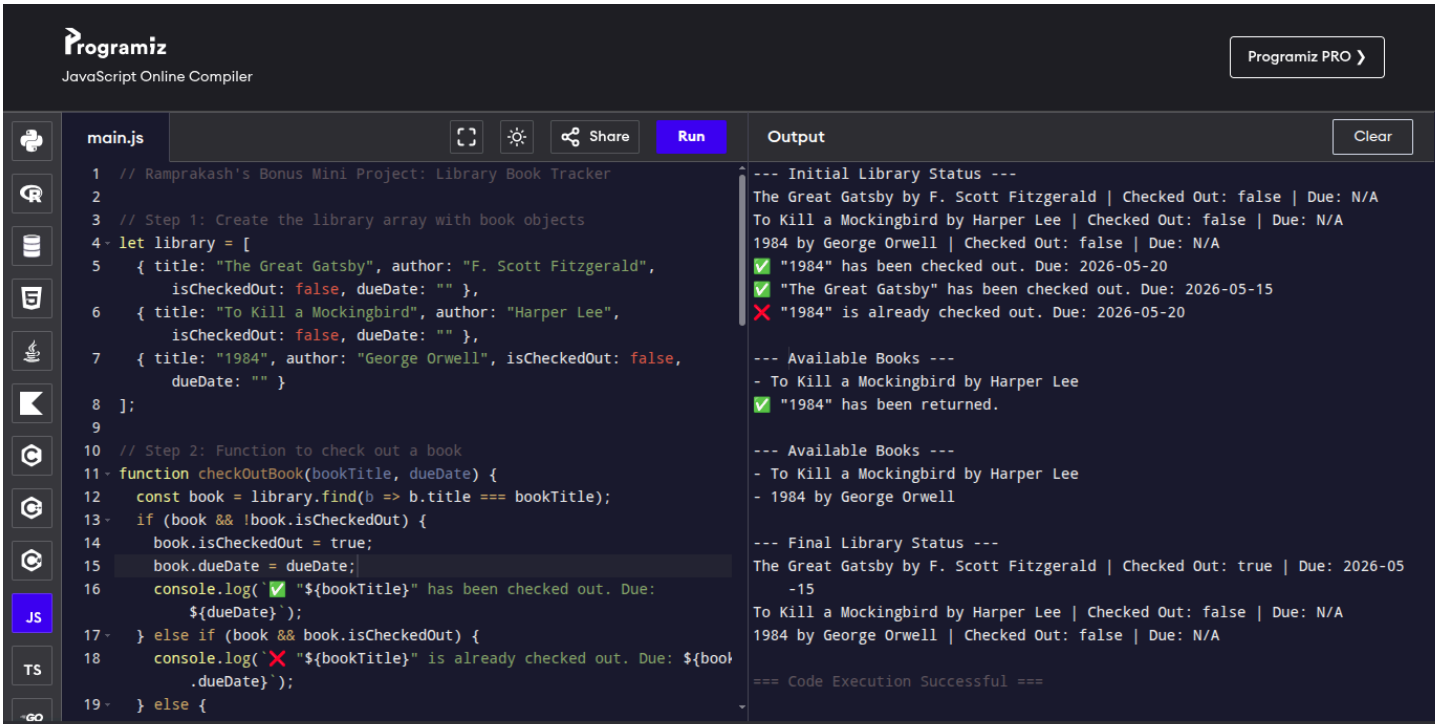
--- Available Books ---
- To Kill a Mockingbird by Harper Lee
✅ "1984" has been returned.

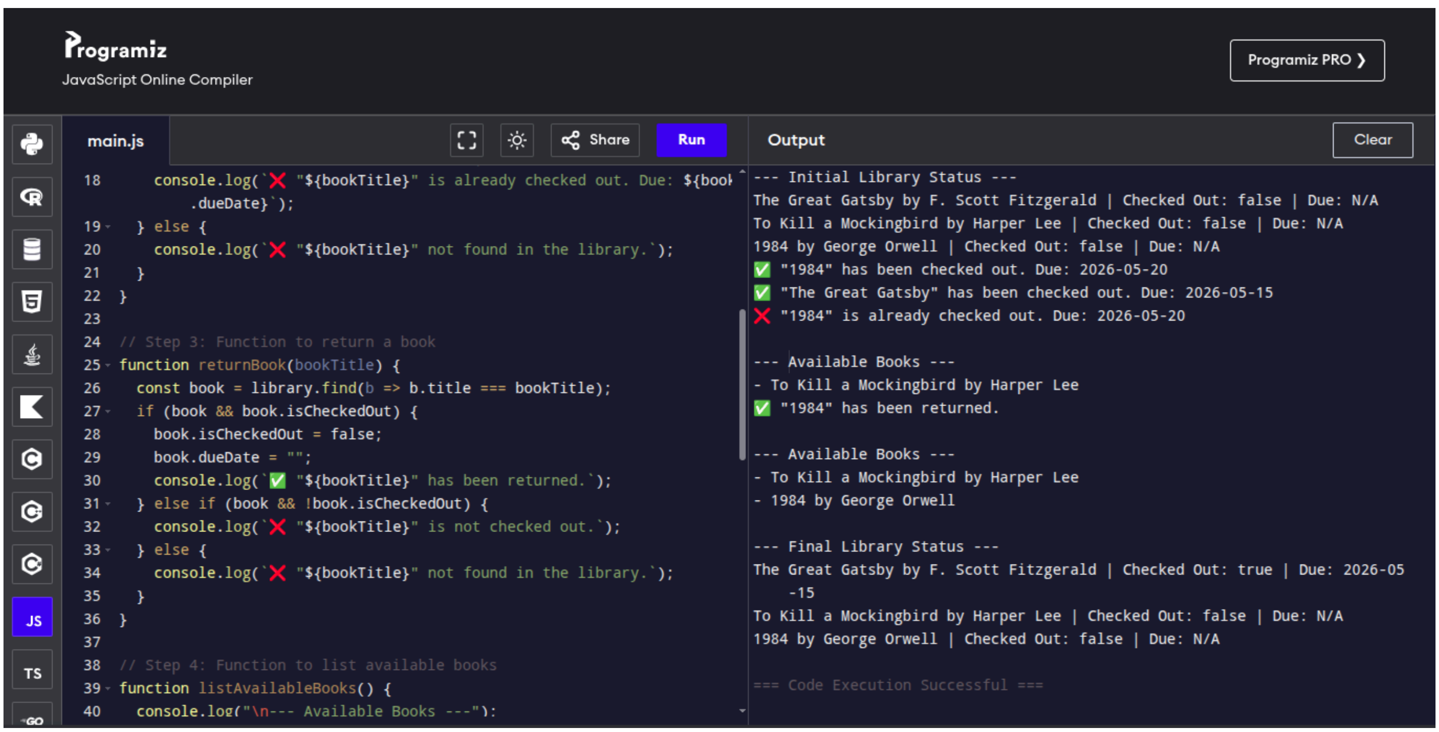
--- Available Books ---
- To Kill a Mockingbird by Harper Lee
- 1984 by George Orwell

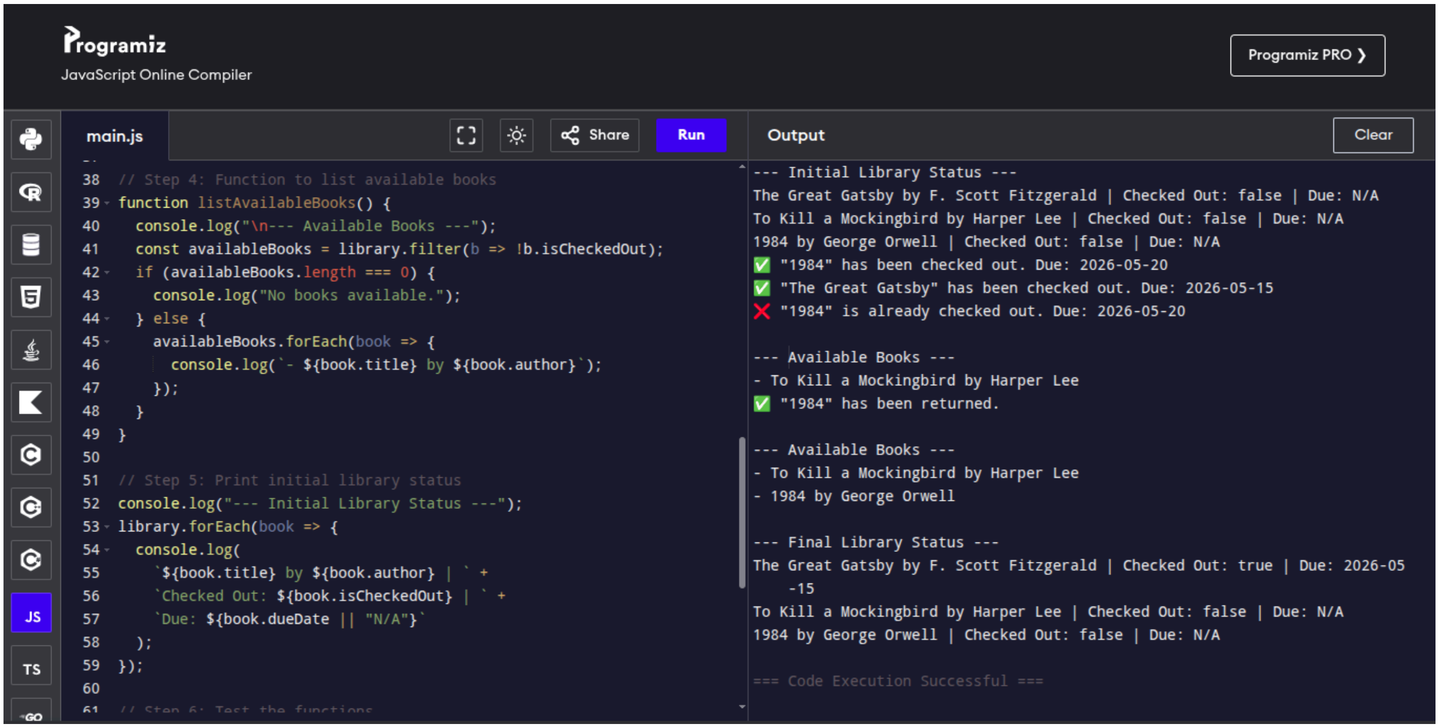
--- Final Library Status ---
The Great Gatsby by F. Scott Fitzgerald | Checked Out: true | Due: 2026-05-15
To Kill a Mockingbird by Harper Lee | Checked Out: false | Due: N/A
1984 by George Orwell | Checked Out: false | Due: N/A

=== Code Execution Successful ===

SCREENSHOTS






Programiz
JavaScript Online Compiler
Programiz PRO ❯
main.js
Share
Run
Output
Clear
// Step 4: Function to list available books
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console.log("\n--- Available Books ---");
const availableBooks = library.filter(b => !b.isCheckedOut);
if (availableBooks.length === 0) {
console.log("No books available.");
} else {
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}
}
// Step 5: Print initial library status
console.log("--- Initial Library Status ---");
library.forEach(book => {
console.log(
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=== Code Execution Successful ===

THANK YOU

Background

BACKGROUND

History with the company

In a paragraph or less, provide an overview of the history with this company. This could include interactions, key events, communications, and milestones.

Key decision-makers

Name	Title	POC	Contact information	Notes
Person	Title	Person	Add information	Add notes
Person	Title	Person	Add information	Add notes
Person	Title	Person	Add information	Add notes

Current product and services

- List the types of products and services this customer prefers
- Preference 2
- Preference 3

Account strategy

ACCOUNT STRATEGY

Indicate how the account strategy plan aligns with the overall goals of the account. Explain how the strategy will be used to pursue specific initiatives to help achieve company goals.

ACTIONS	RESOURCES	METRICS	DUE DATE	OWNER
Add action	Add resources	• Metric 1 • Metric 2	Date	Person Person
Add action	Add resources	• Metric 1 • Metric 2	Date	Person Person
Add action	Add resources	• Metric 1 • Metric 2	Date	Person Person

ACTIONS	RESOURCES	METRICS	DUE DATE	OWNER
Add action	Add resources	• Metric 1 • Metric 2	Date	Person Person
Add action	Add resources	• Metric 1 • Metric 2	Date	Person Person

Competitive and SWOT analysis

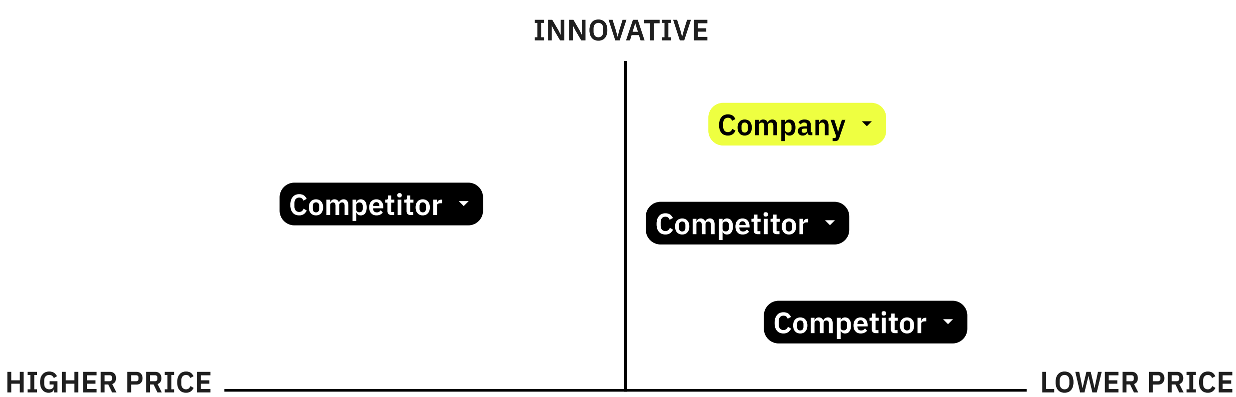
COMPETITIVE ANALYSIS

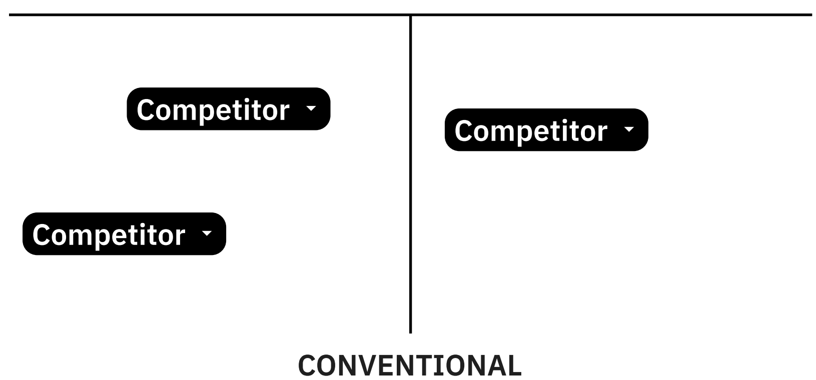
Describe their competitors

- List companies offering similar products or services. Be specific and include large and small competitors.
- Consider alternative products or services that might meet the same customer needs

Summarize their competitive advantage

- Clearly articulate how your client differentiates itself from the competition
- Highlight market opportunities
- Support the competitive advantages with data whenever possible





SWOT ANALYSIS

Strengths	Weaknesses
Internal factors that give the account a competitive advantage.	Internal factors that can hinder the account's progress.
Focus on unique capabilities, resources, or market position.	Identify areas for improvement or potential vulnerabilities.
Examples: Strong brand reputation, innovative products, loyal customer base, efficient operations, skilled workforce.	Examples: Limited market share, outdated technology, high costs, lack of skilled personnel, weak distribution network.

Opportunities	Threats
External factors that the account can use for growth or improvement. Identify emerging trends, market gaps, or favorable conditions. Examples: New market segments, technological advancements, changing regulations, economic growth, competitor weaknesses.	External factors that can negatively impact the account. Identify potential risks and challenges. Examples: Intense competition, economic downturn, changing customer preferences, disruptive technologies, regulatory changes.

Interaction log

INTERACTION LOG

Date	POC	Type of interaction	Notes	Next steps
Date	Person	Meeting	*Interested in new product*	Ongoing
Date	Person	Phone call	Notes	Resolved
Date	Person	Email	Notes	Escalated

ACTION ITEMS

Action item	Assigned to	Deadline	Status
Send customer new catalog	Person	Date	Completed
Action item 2	Person	Date	Completed
Action item 3	Person	Date	Completed

MEETING NOTES

To add a meeting notes building block, type "@meeting notes". Search for or select a meeting to add attendees, notes, and action items.